Young-Sang Lee

Game server programmer · M.S. student

github.com/YSL-RyuDo | ysl-ryudo.github.io

Game server programmer building C++ servers and Unity multiplayer systems. I implement connection/session handling, game-state flows and client integration, with a focus on what must remain consistent across reconnects and asynchronous work. Currently pursuing an M.S. in Games (Engineering) at Hongik University; first author and presenter of two poster studies, and organizer/speaker of the inaugural I2P-CON.

Applied technical skills

- C++ / Windows networking - IOCP, Winsock, sessions, packet dispatch, resource pools and DB worker separation (VAIS, Arcana Duel, Slap Squad).
- C# / Unity - multiplayer client integration, match/profile/shop flows and data-driven gameplay systems.
- Python / experiments - C2PA validation, perceptual-hash retrieval and simulated consumption analysis; test-source and recorded-result review.

Selected project experience

CodeName VAIS

Mar 2026 - Present · 10 members: 5 software, 5 graphics

Server development & Unity multiplayer integration | C++ · Windows IOCP · C# · Unity

- Implemented IOCP connection handling, session management and packet dispatch; connected room/match lifecycle, career/profile, shop and social services to Unity client contracts.
- Separated reconnect presentation recovery from gameplay execution: ReconnectJournal keeps persistent state and excludes replay of damage, rewards and player commands, so restoring the view does not repeat those actions.
- Implemented party/team voice authorization and reconnect-related client flows; integrated Sentinel abilities, lobby, results and player-facing UI with the networked game.

Implementation, contribution evidence and demos

Arcana Duel

Aug 2025 - Feb 2026 · 2 members: 1 programmer, 1 graphics

Programming: C++ server & Unity game integration | C++ · IOCP · DB worker · Unity

- Built the C++ IOCP server with receive/send workers and reusable session, buffer and overlapped-I/O resources; implemented the Unity game/network integration.
- Assigned database requests to a separate DB worker, separating database execution from network completion handling.
- Implemented magic/cooldown, match, profile and rank services, alongside damage/healing and ability integration for the multiplayer magic-combat game.

Implementation, contribution evidence and demos

Slap Squad

Mar 2025 - Feb 2026 · 8 members: 2 programmers, 6 graphics

Server, network integration & shop/profile systems | C++ · Winsock · C# · Unity

- Implemented the C++ Winsock server's account, room, shop and profile flows with connection threads and CSV-backed data; integrated the Unity client with the protocol.
- Handled water-balloon and room-leader updates, purchase UI and the result/reward flow; linked authored client and server commits in the public repositories.
- Developed a CNN study of fixed versus demand-adjusted pricing from the game's item-pricing question; presented it at I2P-CON, then refined the research for the CISC-W 2025 poster.

Implementation, contribution evidence and demos

Education & research experience

2026.09 - Present | M.S. student · Department of Games (Engineering)

Hongik University, Sejong Campus

2021 - 2026.08 | Bachelor's degree · Game Software

Hongik University, Sejong Campus

2025.03 - 2026.08 | Undergraduate researcher · Game Software

Hongik University, Sejong Campus

Research & presentations

WISA 2026

A Verification Method for Game Image Assets Using C2PA Manifest Validation and Perceptual Hash-Based Similarity Analysis

First author & poster presenter, Jeju. Combined C2PA validation with four perceptual hashes; recorded 1,150 validation cases and 4,400 retrieval-evaluation rows. Analyzed both crop tolerance and screenshot-canvas failure conditions; published scripts, results, paper and poster.

CISC-W 2025

Consumption Pattern Analysis of In-Game Pricing Policies Using Convolutional Neural Networks

First author & poster presenter. Simulated 400 players over 30 days, repeated 20 times. Compared fixed pricing with demand-based adjustments; reported 96% CNN classification accuracy for simulated consumption patterns. The result measures pattern classification, not live revenue improvement.

MOD Provenance

Server-Side Multi-Parent Provenance Reconstruction in Heterogeneous Game MOD Packages

Ongoing research combining Java opcode SimHash, image hashes and structured-resource features for multi-parent provenance retrieval. Preserves frozen evaluation baselines and UNKNOWN assignments for unsupported matches.

Leadership & knowledge sharing

2025 inaugural I2P-CON · Operations Team Lead & Speaker

- Recruited the organizing team and led operations for the inaugural student-led game-development conference, jointly hosted by I2P and O2CUBE.
- Presented the CNN pricing-policy research in “Designing Game Economies with Data.” Subsequently refined the research for the CISC-W 2025 poster presentation.
- Produced a conference publication and distributed it to current students and alumni. The official playlist preserves eight talks.

My talk | I2P-CON

O2CUBE · Software Team Lead, 2025

Led C programming and computer architecture study sessions from March to June 2025. Covered C fundamentals through arrays, and guided a semester of computer architecture study.

Additional engineering

DeathGuider: management, facilities and battle-preparation systems. PitchLab: bounded FIFOs and DSP worker separation. GCTLAB: upload validation, storage integration and retry/persistence workflows. Detailed cases are available on the portfolio.